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HOMEWORK 1

1.Sum of two matrices

CODE:

```
#include <stdio.h>
int main(){
   int mat1[10][10], mat2[10][10], rows, cols, i, j;

   printf("Enter size of row:");
   scanf("%d",&rows);

   printf("Enter size of column:");
   scanf("%d",&cols);

   printf("Enter elements in matrix 1:\n");
   for(i=0; i<rows; i++){
      for(j=0; j<cols; j++){
         scanf("%d",&mat1[i][j]);
      }
   }

   printf("Enter elements in matrix 2:\n");
   for(i=0; i<rows; i++){
      for(j=0; j<cols; j++){
         scanf("%d",&mat2[i][j]);
```

```
        }
    }

    printf("\n");
    printf("Matrix 1:\n");
    for(i=0; i<rows; i++){
        for(j=0; j<cols; j++){
            printf("%d ",mat1[i][j]);
        }
        printf("\n");
    }

    printf("\n");
    printf("Matrix 2:\n");
    for(i=0; i<rows; i++){
        for(j=0; j<cols; j++){
            printf("%d ",mat2[i][j]);
        }
        printf("\n");
    }

    printf("\n");
    printf("Sum of two matrices:\n");
    for(i=0; i<rows; i++){
        for(j=0; j<cols; j++){
```

```
            printf("%d ",mat1[i][j] + mat2[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

Enter size of row:2

Enter size of column:2

Enter elements in matrix 1:

1

2

3

4

Enter elements in matrix 2:

5

6

7

8

Matrix 1:

1 2

3 4

Matrix 2:

5 6

7 8

Sum of two matrices:

6 8

10 12

2.Write a C program to read n number of values in an array and display them in reverse order.

CODE:

```
#include <stdio.h>
int main(){
   int arr[100];
   int n,i;

   printf("Enter the size of array:");
   scanf("%d",&n);

   printf("Enter %d elements: \n",n);
   for(i=0; i<n; i++){
      scanf("%d",&arr[i]);
   }
```

```
    printf("Array elements in Reverse order:\n");
    for(i=n-1; i>=0; i--){
        printf("%d ",arr[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter the size of array:5
Enter 5 elements:
12
34
78
65
98
Array elements in Reverse order:
98 65 78 34 12
```

3.Count the total number of duplicate elements in an array.

CODE:

```
#include <stdio.h>
int main(){
```

```
int arr[100], n, i, j,dup_count = 0;


printf("Enter the size of array:");
scanf("%d",&n);


printf("Enter %d elements: \n",n);
for(i=0; i<n; i++){
   scanf("%d",&arr[i]);
}


   for(i = 0; i < n; i++) {
   for(j = 0; j < i; j++) {
      if(arr[i] == arr[j]) {
         dup_count++;
         break;
      }
   }
}
   printf("Total duplicate elements in array: %d\n",dup_count);


 return 0;
}
```

OUTPUT:

Enter the size of array:5

Enter 5 elements:

11

12

11

43

12

Total duplicate elements in array: 2

4.Print unique elements in an array.

CODE:

```
#include <stdio.h>
int main(){
   int arr[100], n, i, j, count;

   printf("Enter the size of array:");
   scanf("%d",&n);

   printf("Enter %d elements: \n",n);
   for(i=0; i<n; i++){
      scanf("%d",&arr[i]);
   }

   printf("Unique elements in the array: ");
   for(i=0; i<n; i++){
      count=0;
```

```
        for(j=0; j<n; j++){
          if(arr[j] == arr[i]){
            count++;
          }
        }
        if(count == 1) {
          printf("%d ", arr[i]);
        }
      }

      return 0;
    }
```

OUTPUT:

Enter the size of array:8

Enter 8 elements:

12

34

56

12

11

12

10

68

Unique elements in the array: 34 56 11 10 68

5.Insert in sorted array.

CODE:

```
#include <stdio.h>
int main(){
   int arr[100],n,i,val;

   printf("Enter the size of sorted array:");
   scanf("%d",&n);

   printf("Enter %d elements: \n",n);
   for(i=0; i<n; i++){
      scanf("%d",&arr[i]);
   }

   printf("Enter the value:");
   scanf("%d",&val);

   for(i=n-1; i>=0 && arr[i]>val; i--){
      arr[i+1]=arr[i];
   }

   arr[i+1] = val;
   n++;
```

```
    printf("New array: \n",n);
    for(i=0; i<n; i++){
        printf("%d ",arr[i]);
    }

    return 0;
}
```

OUTPUT:

Enter the size of sorted array:5

Enter 5 elements:

12

45

56

67

89

Enter the value:34

New array:

12 34 45 56 67 89

6.Insert in sorted array.

CODE:

```
#include <stdio.h>
int main() {
    int arr[100], n, i, pos, value, temp;
```

```
printf("Enter the size of array:");
scanf("%d",&n);

printf("Enter the elements:");
for (i=0; i<n; i++){
   scanf("%d",&arr[i]);
}

printf("Enter the position to insert:");
scanf("%d",&pos);

printf("Enter the value:");
scanf("%d",&value);

i =n - 1;
while( i >= pos-1){
   arr[i+1] = arr[i];
   i--;
}
arr[pos-1] = value;
n = n+1;

printf("New array:\n");
for (i=0; i<n; i++){
```

```
        printf("%d ",arr[i]);
    }

return 0;
}
```

OUTPUT:

Enter the size of array:5

Enter the elements:12

34

56

78

90

Enter the position to insert:3

Enter the value:64

New array:

12 34 64 56 78 90

7.Matrix multiplication.

CODE:

```
#include <stdio.h>
int main() {
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows of first matrix: ");
```

```
scanf("%d", &r1);


 printf("Enter columns of first matrix: ");
scanf("%d", &c1);


printf("Enter rows second matrix: ");
scanf("%d", &r2);


printf("Enter columns of first matrix: ");
scanf("%d", &c2);


if (c1 != r2) {
    printf("Matrix multiplication is not possible.\n");
    return 0;
}


int A[r1][c1], B[r2][c2], C[r1][c2];


printf("Enter elements of first matrix:\n");
for (i = 0; i < r1; i++)
    for (j = 0; j < c1; j++)
        scanf("%d", &A[i][j]);
printf("Enter elements of second matrix:\n");
for (i = 0; i < r2; i++)
    for (j = 0; j < c2; j++)
```

```
            scanf("%d", &B[i][j]);
    for (i = 0; i < r1; i++)
        for (j = 0; j < c2; j++)
            C[i][j] = 0;

    for (i = 0; i < r1; i++) {
        for (j = 0; j < c2; j++) {
            for (k = 0; k < c1; k++) {
                C[i][j] += A[i][k] * B[k][j];
            }
        }
    }
    printf("Resultant Matrix:\n");
    for (i = 0; i < r1; i++) {
        for (j = 0; j < c2; j++) {
            printf("%d ", C[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

Enter rows of first matrix: 2

Enter columns of first matrix: 2

Enter rows second matrix: 2

Enter columns of first matrix: 2

Enter elements of first matrix:

1

2

3

4

Enter elements of second matrix:

5

6

7

8

Resultant Matrix:

19 22

43 50

8.Matrix transpose.

CODE:

```
#include <stdio.h>

int main(){

   int arr[10][10], rows, cols,i, j;

    printf("Enter size of row:");

   scanf("%d",&rows);
```

```
printf("Enter size of column:");
scanf("%d",&cols);

printf("Enter elements of the matrix:\n");
for(i=0; i<rows; i++){
   for(j=0; j<cols; j++){
      scanf("%d",&arr[i][j]);
   }
}

printf("\n");
printf("Matrix:\n");
for(i=0; i<rows; i++){
   for(j=0; j<cols; j++){
      printf("%d ",arr[i][j]);
   }
   printf("\n");
}

printf("Transpose of the matrix:\n");
for(i=0; i<cols; i++){
   for(j=0; j<rows; j++){
      printf(" %d ",arr[j][i]);
   }
   printf("\n");
```

```
    }

    return 0;
}
```

OUTPUT:

```
Enter size of row:2
Enter size of column:3
Enter elements of the matrix:
12
34
45
56
67
78

Matrix:
12 34 45
56 67 78
Transpose of the matrix:
 12  56
 34  67
 45  78
```

9.Right diagonal sum.

CODE:

```
#include <stdio.h>
int main(){
   int arr[10][10], n, i, j, sum=0;

   printf("Enter size of matrix:");
   scanf("%d",&n);

   printf("Enter elements of the matrix:\n");
   for(i=0; i<n; i++){
      for(j=0; j<n; j++){
         scanf("%d",&arr[i][j]);
      }
   }

   printf("\n");
   printf("Matrix:\n");
   for(i=0; i<n; i++){
      for(j=0; j<n; j++){
         printf("%d ",arr[i][j]);
      }
      printf("\n");
   }

   for(i = 0; i < n; i++) {
```

```
        for(j = 0; j < n; j++) {
            if(i + j == n - 1) {
                sum += arr[i][j];
            }
        }
    }

    printf("\nRight diagonal sum = %d", sum);

    return 0;
}
```

OUTPUT:

```
Enter size of matrix:3
Enter elements of the matrix:
10
11
12
23
45
76
34
14
15

Matrix:
```

10 11 12

23 45 76

34 14 15

Right diagonal sum = 91

10.Check for identity matrix.

CODE:

```
#include <stdio.h>
int main(){
 int arr[10][10], n, i, j,found=1;

  printf("Enter size of matrix:");
  scanf("%d",&n);

  printf("Enter elements of the matrix:\n");
  for(i=0; i<n; i++){
    for(j=0; j<n; j++){
      scanf("%d",&arr[i][j]);
    }
  }

  printf("\n");
  printf("Matrix:\n");
```

```
    for(i=0; i<n; i++){
      for(j=0; j<n; j++){
        printf("%d ",arr[i][j]);
      }
      printf("\n");
    }

    for(i=0; i<n; i++){
      for(j=0; j<n; j++){
        if(i == j){
          if(arr[i][j] !=1)
          found =0;
        }
        else{
           if(arr[i][j] != 0){
          found = 0;
        }
      }

    }
  }
    if(found==1){
      printf("Identity matrix");
    }
    else{
```

```
        printf("Not an identity matrix");
    }
    return 0;
}
```

OUTPUT:

Enter size of matrix:3

Enter elements of the matrix:

1

0

0

0

1

0

0

0

1

Matrix:

1 0 0

0 1 0

0 0 1

Identity matrix

11.To find out number of vowels, consonants, digits, and white spaces.

CODE:

```
#include <stdio.h>
int main(){
   char str[100];
   int i;
   int digits =0, vowels=0, con=0, spaces=0;
   printf("Enter the string: ");
   fgets(str, sizeof(str), stdin);

   for(i=0; str[i] != '\0'; i++){
      if(str[i] >='0' && str[i] <='9'){
         digits++;
      }
      else if(str[i] == 'a' ||str[i] == 'e' ||str[i] == 'i' ||str[i] == 'o' ||str[i] == 'u' ||str[i]
== 'A' ||str[i] == 'E' ||str[i] == 'I' ||
      str[i] == 'O' ||str[i] == 'U'){
         vowels++;
      }
      else if(str[i] == ' '){
         spaces++;
      }
      else if((str[i] >= 'A' && str[i] <= 'Z') ||
            (str[i] >= 'a' && str[i] <= 'z')) {
         con++;
```

```
        }
    }
    printf("Digits: %d\nVowels: %d\nConsonants: %d\nWhite spaces:
%d",digits, vowels, con, spaces);
    return 0;
}
```

OUTPUT:

Enter the string: I am very much interested in Problem solving 5028

Digits: 4

Vowels: 13

Consonants: 24

White spaces: 8

12.Frequency of character in a string.

CODE:

```
#include <stdio.h>
int main(){
    char str[100];
    int i, freq[26] = {0};

    printf("Enter the string: ");
    fgets(str,sizeof(str), stdin);

    for(int i =0; str[i] !='\0'; i++){
        if(str[i] >= 'A' && str[i] <= 'Z' ){
```

```
            str[i] = str[i] + 32;
        }
    }
    for(int i =0; str[i] !='\0'; i++){
        freq[str[i] - 'a']++;
}
    for (i = 0; i < 26; i++) {
        if (freq[i] > 0){
            printf("%c = %d\n", i + 'a', freq[i]);
        }
    }
    return 0;
}
```

OUTPUT:

Enter the string: programming

a = 1

g = 2

i = 1

m = 2

n = 1

o = 1

p = 1

r = 2

13.Count words in a string.

CODE:

```
#include <stdio.h>
int main(){
    char sen[100];
    int i,words =1;
    printf("Enter the sentence:");
    fgets(sen,sizeof(sen),stdin);
    if(sen[0] == '\0' || sen[0] == '\n'){
      words = 0;
    }
     for( i = 0; sen[i] != '\0'; i++){
      if(sen[i] ==' ' && sen[i+1] != ' ' && sen[i+1] != '\0'){
          words++;
      }
}
  printf("Number of words in the given sentence:%d",words);
  return 0;
}
```

OUTPUT:

Enter the sentence:I am very much interested in Problem solving

Number of words in the given sentence:8

14.Sort string array.

CODE:

```
#include <stdio.h>
#include <string.h>

int main() {
   char str[10][50], temp[50];
   int n, i, j;

   printf("Enter the number of strings: ");
   scanf("%d", &n);

   getchar();

   printf("Enter the strings:\n");
   for(i = 0; i < n; i++) {
      fgets(str[i], sizeof(str[i]), stdin);
      str[i][strcspn(str[i], "\n")] = '\0';
   }

   for(i = 0; i < n - 1; i++) {
      for(j = i + 1; j < n; j++) {
         if(strcmp(str[i], str[j]) > 0) {
            strcpy(temp, str[i]);
            strcpy(str[i], str[j]);
            strcpy(str[j], temp);
```

```
            }
        }
    }

    printf("\nSorted strings:\n");
    for(i = 0; i < n; i++) {
        printf("%s\n", str[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter the number of strings: 3
Enter the strings:
Mango
Grapes
Orange

Sorted strings:
Grapes
Mango
Orange
```

15.Toggle case of a sentence.

CODE:

```
#include <stdio.h>
int main(){
   char str[100];
   printf("Enter string:");
   fgets(str,sizeof(str),stdin);

   for(int i = 0; str[i] != '\0'; i++){
      if(str[i] >= 'A' && str[i] <= 'Z'){
         str[i] = str[i] + 32;
      }
      else if(str[i] >= 'a' && str[i] <= 'z'){
         str[i] = str[i] - 32;
   }
}
printf("Case changed string:%s",str);
return 0;
}
```

OUTPUT:

Enter string:HELlo woRLd

Case changed string:helLO WOrlD